

Assigning Representative Field Attribute to Regrow Fields

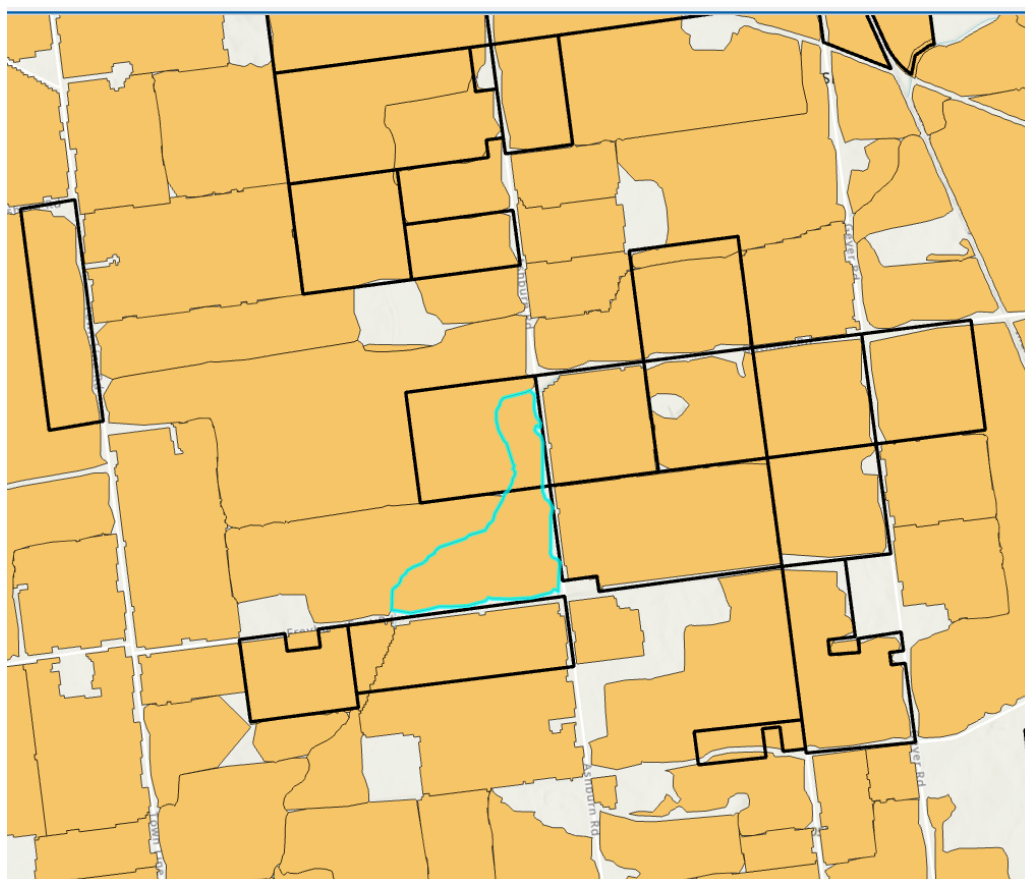
Supporting Documentation

In the DISES survey, one block of questions asks farmers about the characteristics of their “**representative**” field. Because the survey does not specify which field each farmer considers representative, we identify a candidate representative field from the corresponding Regrow fields by spatially overlaying Regrow and DISES geometries and comparing attributes available in both datasets: **field area** and **main crop cultivated in 2023**.

When merging the Regrow and DISES datasets, we retain all spatial overlaps and create **merge-quality indicators**. As the sources of Regrow field boundaries and DISES parcel boundaries differ, the overlapping area may be relatively small in some cases. Below are examples where it is unclear whether a Regrow field should be considered a valid candidate for a representative field.

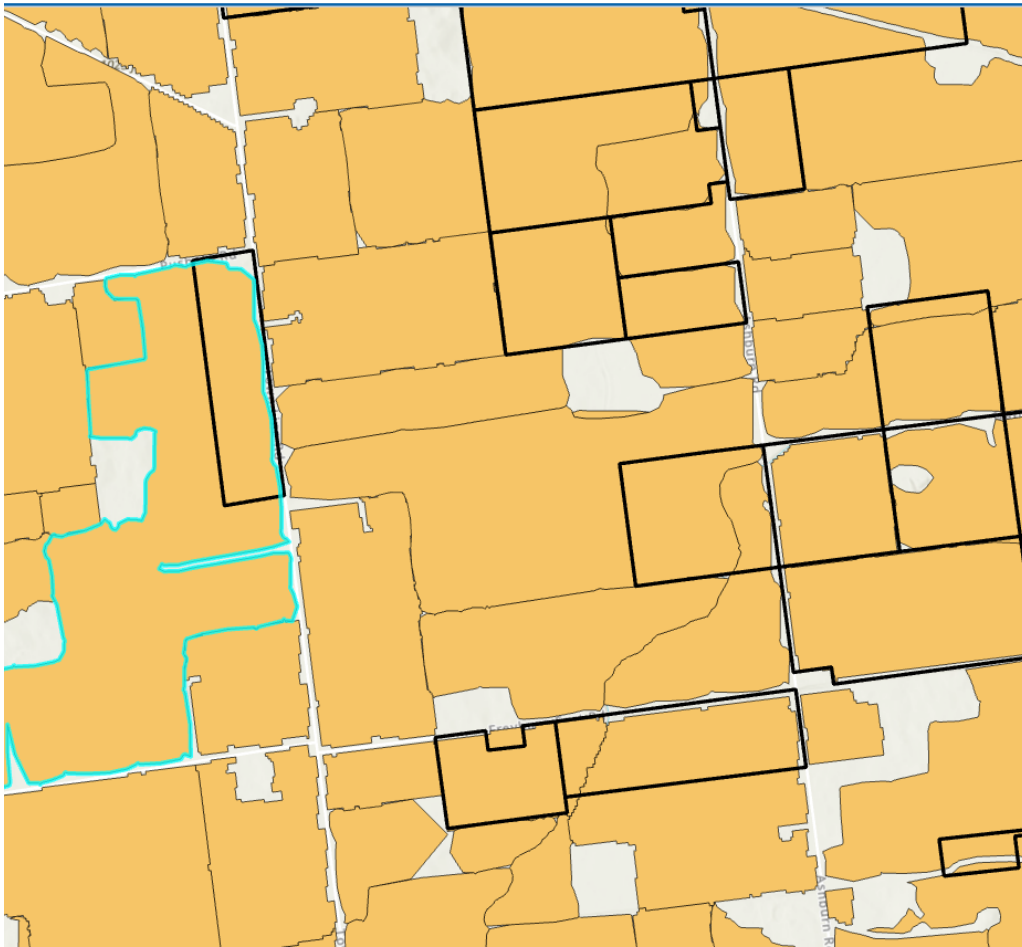
Ambiguous Overlap Cases

Case 1. A Regrow field has only a small overlap with a DISES parcel



Note: Regrow fields are shown as orange polygons; the examined Regrow polygon is highlighted in blue; DISES parcels are outlined in black.

Case 2. A DISES parcel lies entirely within a much larger Regrow field



Note: Regrow fields are shown as orange polygons; the examined Regrow polygon is highlighted in blue; DISES parcels are outlined in black.

In both cases, it is unclear whether the Regrow field can reliably represent the farmer's representative field. To address this uncertainty, we adopt a **conservative approach** and exclude such cases by applying a threshold to the overlap metric.

Filter applied:

- $\text{overlap_area_share_dises} > 50\%$ (applied to the entire joined dataset)
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Classification of Candidate Representative Fields

Candidate Regrow fields are classified into four levels based on their likelihood of being the representative field. To get the sample of all representative fields, it is necessary to combine Regrow fields across all levels.

Level 1

Explanation:

Regrow fields assigned to **Level 1** have the highest likelihood of being the representative field (RF) for a farmer with a given comp_id_dises. Only Regrow fields that **match both RF size and main crop** answers are included. In addition, the confidence score for the Regrow main crop in 2023 must exceed **75%**, as recommended by the Regrow vendor.

Filter:

- match_quality_dises = "A" AND crop_conf_23_{cycle number} > 75%

*(if multiple Regrow fields qualify, select the field with the **most similar area** to the representative field)*

Level 2

Explanation:

Regrow fields assigned to **Level 2** have a medium likelihood of being the representative field. A Regrow field is assigned to this level if it satisfies **at least one** of the following conditions:

- Matches both RF size and main crop answers, but the Regrow crop confidence score for 2023 is **below 75%**.
- Matches only RF size answer, while the RF main crop answer is missing in the DISES survey.
- Matches only RF main crop answer, while the RF size answer is missing in the DISES survey, and the Regrow crop confidence score for 2023 is **above 75%**.

Filter:

- match_quality_dises = "A" AND crop_conf_23_{cycle number} < 75%

(If multiple fields qualify, select the one with the most similar area)

- match_quality_dises = "B_area" AND field_crop_23_dises is missing

(If multiple fields qualify, select the one with the most similar area)

- match_quality_dises = "B_crop" AND field_size_dises is missing AND crop_conf_23_{cycle number} > 75%

(If multiple fields qualify, select the one with the largest overlap_area_share_dises)

Level 3

Explanation:

Regrow fields assigned to **Level 3** have a low-to-medium likelihood of being the representative field. A Regrow field is assigned to this level if it satisfies **at least one** of the following conditions:

- Matches only RF main crop answer, RF size answer is missing in the DISES survey, and the Regrow crop confidence score for 2023 is **below 75%**.
- Matches RF size answer but does not match RF main crop answer; however, the Regrow crop confidence score for 2023 is **below 75%**, suggesting a potential crop classification error.

Filter:

- match_quality_dises = "B_crop" AND field_size_dises is missing AND crop_conf_23_{cycle number} < 75%

(If multiple fields qualify, select the one with the largest overlap_area_share_dises)

- match_quality_dises = "B_area" AND field_crop_23_dises does not match AND crop_conf_23_{cycle number} < 75%

(If multiple fields qualify, select the one with the most similar area)

Level 4

Explanation:

Regrow fields assigned to **Level 4** have a low likelihood of being the representative field. A Regrow field is assigned to this level if it satisfies **at least one** of the following conditions:

- RF size answer is missing in the DISES survey and the Regrow main crop does not match the DISES RF main crop, while the Regrow crop confidence score for 2023 is **below 75%**, indicating possible crop misclassification.
- Both RF size and crop answers are missing in the DISES survey but only a single parcel is assigned to a farmer in DISES dataset.

Filter:

- field_size_dises is missing AND field_crop_23_dises does not match AND crop_conf_23_{cycle number} < 75%

(If multiple fields qualify, select the one with the largest overlap_area_share_dises)

- field_crop_23_dises is missing AND field_size_dises is missing AND n_parcels_dises = 1

(If multiple fields qualify, select the one with the largest overlap_area_share_dises)